

CS707 – Generative Models for NLP and Multimodal Reasoning

2026-27 Term 1

Week 1: Introduction to Generative Models

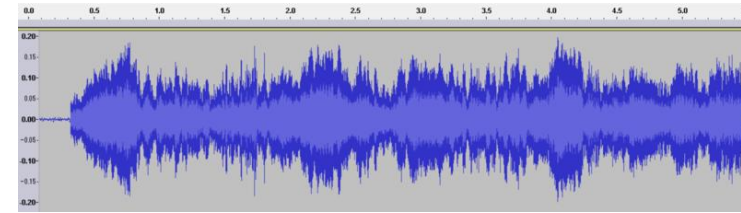
Dr. Wei Gao
Office: SCIS-1/4057
Email: weigao@smu.edu.sg
Tel.: 68261369

Introduction

Challenge: Understand complex, unstructured inputs



Computer Vision



Computational Speech

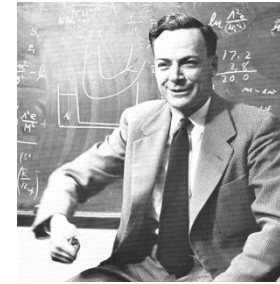


Natural Language Processing



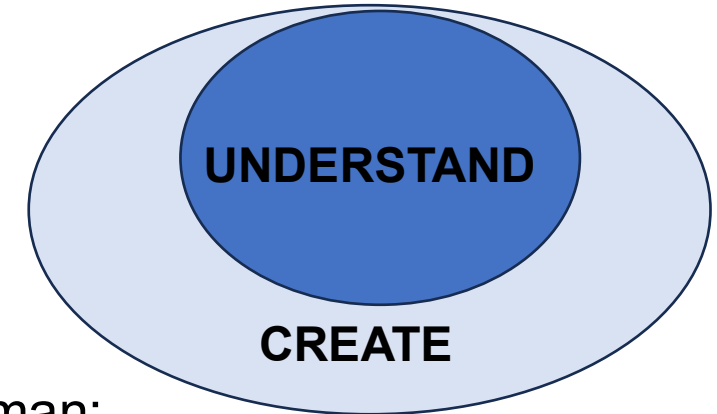
Robotics

Understanding vs. Creation (Generation)



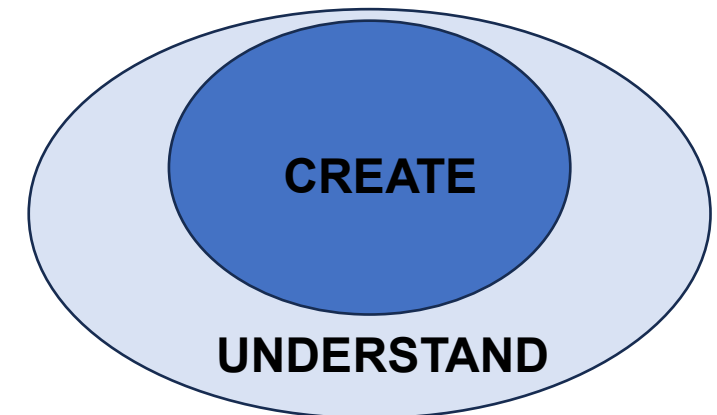
Richard Feynman:

“What I cannot create, I do not understand”



Generative AI: *“What I understand, I can **create**, and even beyond”*

Trustworthy/Responsible AI: *“If I can **create** it, I must **understand** it”*



Outline

- Logistics
- Generative Models Overview
- Roadmaps and Challenges
- Recent Advances

Learning Objectives

This course prepares students for cutting-edge research and AI applications by providing both theoretical knowledge and hands-on experience in generative AI.

- (1) Understand generative AI fundamentals – explain core generative modelling techniques including transformers, VAEs, GANs, and diffusion models, and their applications in NLP and multimodal AI.
- (2) Develop and adapt LLMs and multimodal models (e.g., GPT, T5, LLaMA, CLIP) for downstream tasks, applying techniques such as prompt tuning, fine-tuning, and retrieval-augmented generation (RAG).
- (3) Evaluate model quality using quantitative metrics and qualitative evaluation, and address challenges in bias, fairness, and efficiency.
- (4) Design, implement, and present a research-driven project demonstrating the use of generative models in NLP and multimodal reasoning.

Our TA

BHOSALE Rimmon Saloman

rs.bhosale.2024@phdcs.smu.edu.sg

What do TA do for you?

- Give you tutorials
- Help you with hands-on labs
- Provide consultation online and offline
- Answer your questions on Piazza:
Join in using your SMU email from <https://piazza.com/smu.edu.sg/fall2026/cs707>
- Help in groups formation and logistics

Course Delivery

This course will be delivered via:

- Seminar style lectures
- Tutorials on specific and ancillary topics for replenishing lectures
- Practical labs for training hands-on skills
- Paper presentation and self-exploration

Section	Day	Time	Location
G1	Tuesday	7:00-10:15pm	SCIS-1 Seminar Rm 2-4 (Week 1-12) SOE/SCIS-2 Seminar Rm 4-1 (Week 13)

Structure of each session (except for weeks 1, 2, 7, 8, 13):

- 60-min lecture
- 30-min tutorial / lab
- 90-min student presentations + discussions (2-3 group presentations)

Weekly Schedule

Week/Day	Lecture	Tutorial	Presentation	Remark
1 (20 Aug)	Introduction to Generative AI	-	-	Presentation groups formation Project guideline released
2 (27 Aug)	Probabilistic & Neural Generative Models	Tutorial 1	-	
3 (03 Sep)	Transformers and Sequence Modeling	Tutorial 2	Presentation-1	
4 (10 Sept)	Mixture of Experts	Tutorial 3	Presentation-2	
5 (17 Sept)	Large Language Models (I)	-	Presentation-3	In-class Quiz 1
6 (24 Sept)	Large Language Models (II)	Tutorial 4	Presentation-4	
7 (01 Oct)	Mid-term Project Presentation	-		
8 (08 Oct)	Recess week (no class)	-		Project proposal due Assignment released
9 (15 Oct)	Multimodal Learning and Representation	-	Presentation-5	In-class Quiz 2
10 (22 Oct)	Diffusion and Hybrid Models for Multimodal AI	Tutorial 5	Presentation-6	
11 (29 Oct)	Knowledge-Augmented Generation	Tutorial 6	Presentation-7	
12 (05 Nov)	LLM Reasoning	Tutorial 7	Presentation-8	In-class Quiz 3
13 (12 Nov)	Final Project Presentation	-		
14	Study week (no class)	-		Assignment due
15	Exam week (no class)	-		Project report due

All due times are 23:59 Fridays of the specific weeks

Assessment

Item		Description	Weight	Due/test date and time
Class participation		Individual attendance + engagement	10%	Every week
Quiz 1		Individual assessment based on the performance on answering mini quiz questions.	5%	Week 5 (in class)
Quiz 2			5%	Week 9 (in class)
Quiz 3			10%	Week 12 (in class)
Assignment		Individual assessment based on the exhibited competency with materials introduced in the classes	10%	Week 14 (11:59pm, Friday)
Paper Presentation		Individual assessment based on paper presentation and discussion	20%	Week 2 onward (in class)
Research Project	Mid-term presentation + Proposal	Group assessment based on a presentation and written proposal reporting the project’s goals and plan	10%	Week 7 (11:59pm, Friday) - proposal
	Final Presentation	Group assessment based on the presentation quality	10%	Week 13 (in class)
	Report	Group assessment based on a group written report	15%	Week 15 (11:59pm, Friday)
	Peer review	Individual assessment based on peer evaluation given by group members	5%	Week 15 (11:59pm, Friday)

Paper Presentation (presenter)

- **Format:**

- Physical presentation in class
- **Individual presentation** (no grouping)
- 30 minutes per presentation including discussion/QA
- Three presentations each week

Check this master file for weekly presentation schedule:

<https://docs.google.com/spreadsheets/d/1lLqzqR7mZ1okc2OiPIGXagmw99gchlW2PTffnw2BU3k/edit?usp=sharing>

- **Notes:**

- Presentation and discussion quality will be assessed (**10%**)
- Paper will be provided to presenter **one week before class (column F)**
- Email TA the URL of your slides **before class** (TA will fill **column G**)
- Ask audience **1-3** questions on the first slide, and answer audience questions

Paper Presentation (audience)

- **Format:**

- Write your comments/questions to the presenter and your answers for the questions raised by the presenter while listening, and submit the Google form following the link given in **column H**:

<https://docs.google.com/spreadsheets/d/1ILqzqR7mZ1okc2OiPIGXagmw99gchlW2PTffnw2BU3k/edit?usp=sharing>

- **Notes:**

- Comment/question/answer quality will be assessed (10%)
- Reading paper beforehand helps you better understand the work and prepared for asking/answering questions
- Comments/questions/answers submitted by **the end of day**

Quiz

- **Format:**
 - Physical (eLearn with lockdown browser)
 - Close book, close notes
 - 30-60 minutes
 - Short answer questions
- **Coverage:**
 - A research paper relevant to the topics up to the quiz week
- **NO make-up quizzes**
 - We generally do NOT offer make-ups for **missed quizzes** – except you're sick (with medical certificate)

Assignment and Project

Item	Topic	Remarks
Assignment* (10%)	Novel multimodal model design and justification. Specific assignment questions will be announced in due course.	Questions release in Week 7 ; due in Week 14
Research Project** (40%)	Project themes should be self-defined and research-oriented. You may pursue one of three directions: 1) Address a well-defined problem by developing a substantively novel model or methodology, with rigorous comparison against strong SoTA baselines. 2) Identify an important research question that is inadequately captured by existing tasks, formulate a new task or setting, and develop and validate an appropriate solution. 3) Develop a theoretically motivated learning objective, scaling or efficiency strategy, or new evaluation methodology/protocol that addresses an important limitation of existing approaches. The expected standard is research that could plausibly be developed toward submission to a top conference – your work will be evaluated against criteria of A*/A-level conferences.	Guideline release in Week 1-2 ; final due in Week 15

* The assignment is an **individual work** task – No mutual discussion.

** The project is a **group work** task. Each group will submit a proposal and a final report describing the project (detailing individual's main contribution should be provided). Each member is required to participate in the presentation. There will be a peer evaluation component in the project.

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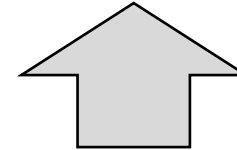
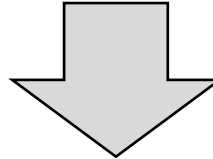
Generative Modeling: Computer Graphics

How to generate natural images with a computer?

High level
description

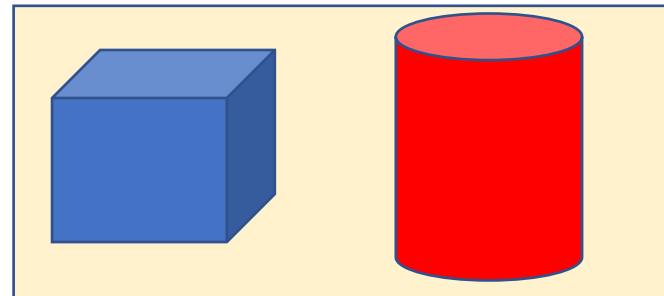
```
Cube(color=blue, position=(x,y,z), size=...)  
Cylinder(color=red, position=(x',y',z'), size=..)
```

Generation (graphics)



Inference (vision as
inverse graphics)

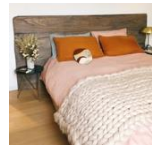
Raw sensory
outputs



Many of our models will have **similar structure (generation + inference)**

Statistical Generative Models

Statistical generative models are **learned from data**



...



+



Data
(e.g., images of bedrooms)

Prior Knowledge
(e.g., physics, materials, ..)

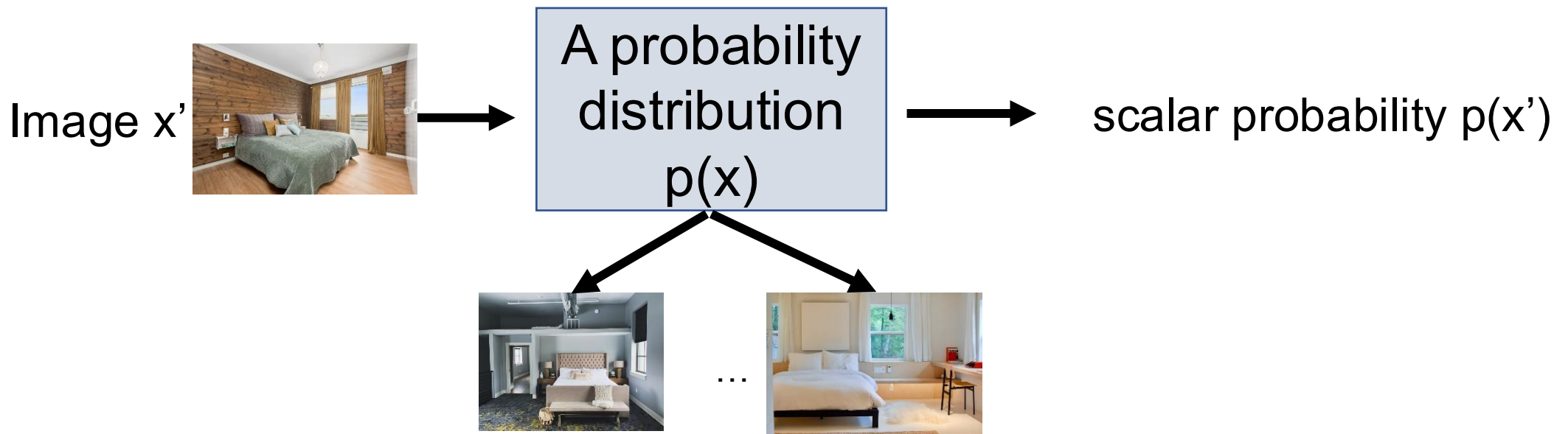
Priors are always necessary, but there is a spectrum



Statistical Generative Models

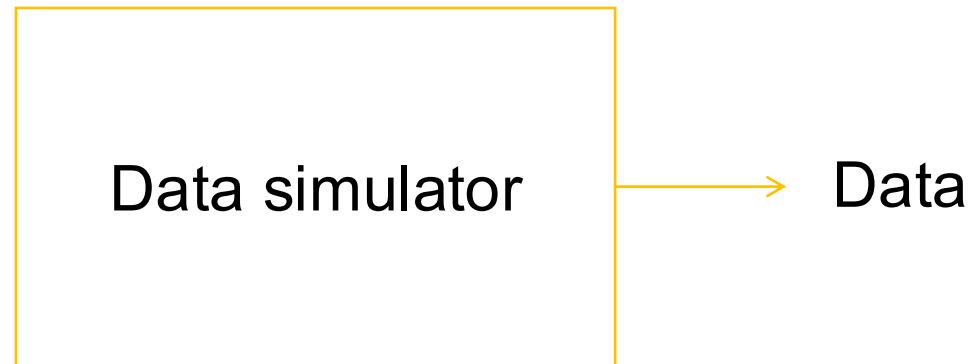
A statistical generative model is a **probability distribution** $p(x)$

- **Data:** samples (e.g., images of bedrooms)
- **Prior knowledge:** parametric form (e.g., Gaussian?), loss function (e.g., maximum likelihood?), optimization algorithm, etc.

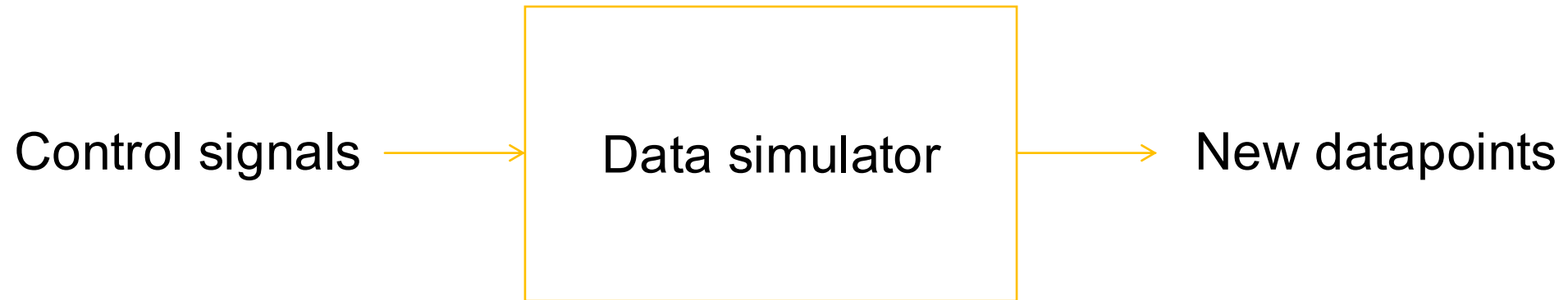


It is generative because **sampling from $p(x)$ generates new images**

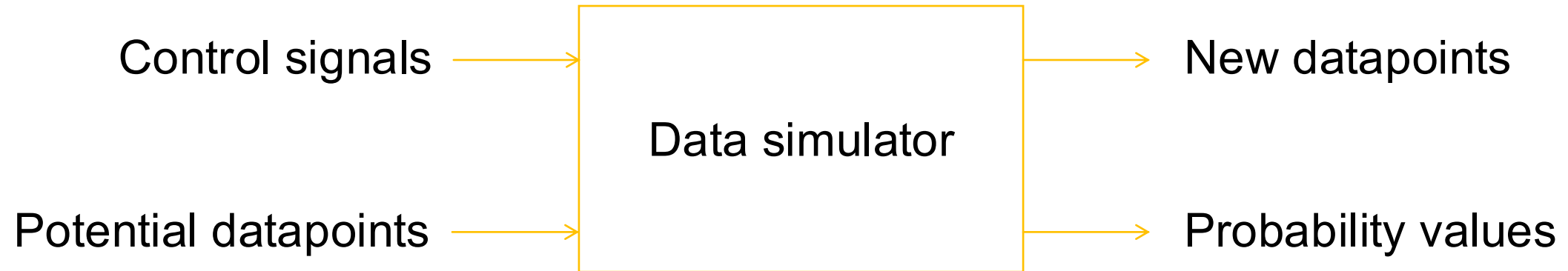
Building a simulator for the data generating process



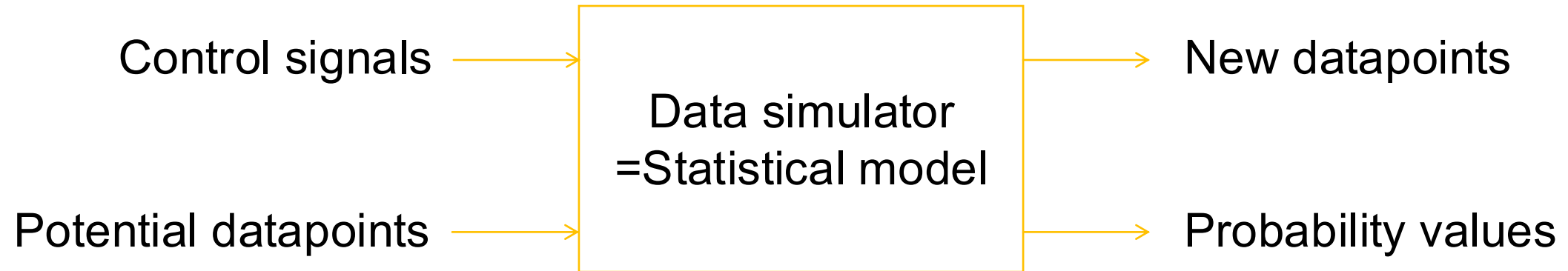
Building a simulator for the data generating process



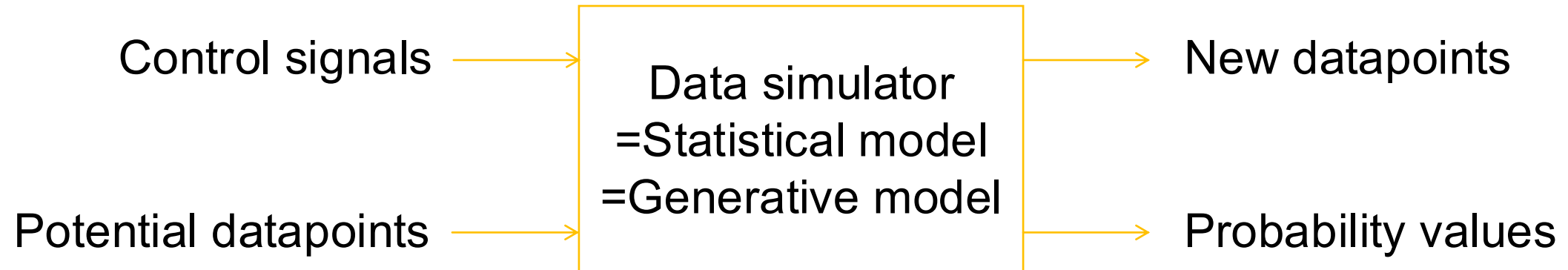
Building a simulator for the data generating process



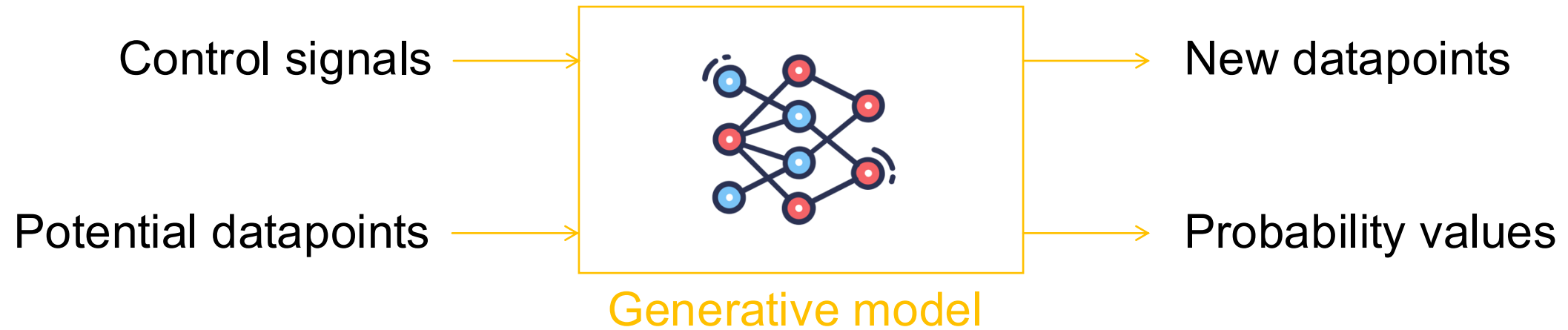
Building a simulator for the data generating process



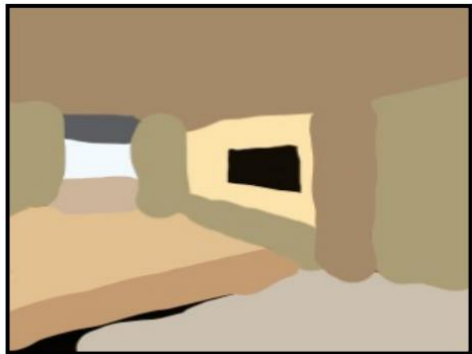
Building a simulator for the data generating process



Building a simulator for the data generating process



Data generation in the real world



Generative model
of realistic images



“Ace of Pentacles”



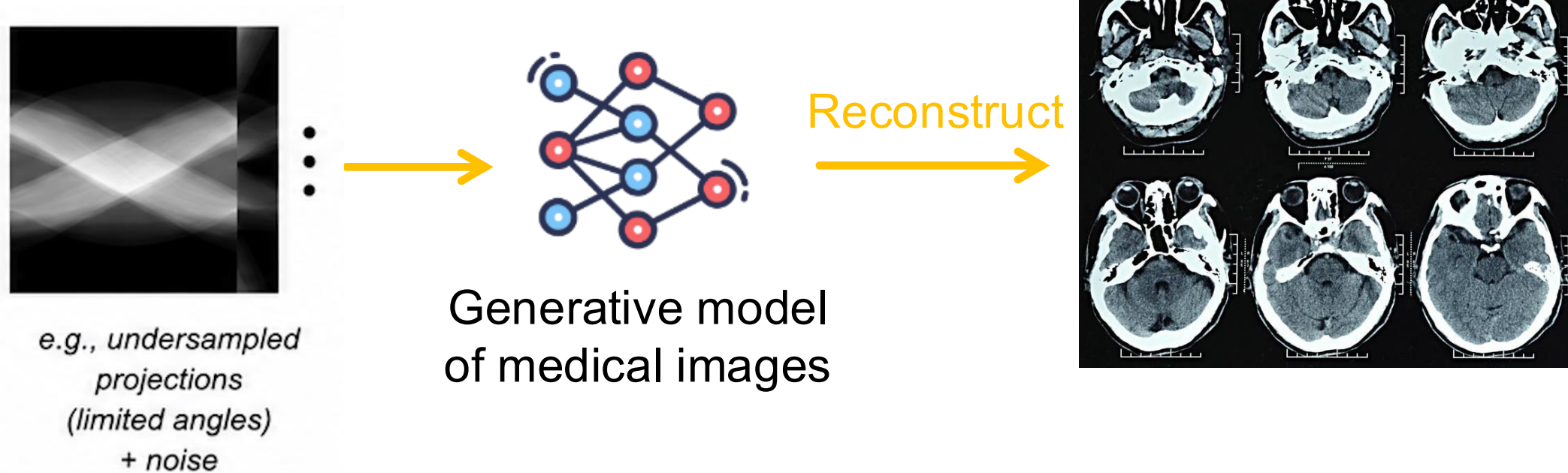
Generative model
of paintings



Language-guided artwork creation

<https://chainbreakers.kath.io> @RiversHaveWings

Solving inverse problems with generative models



Medical image reconstruction

[Song et al., ICLR 2022]

Outlier detection with generative models



High
probability
→



Generative model
of traffic signs

←
Low
probability



Outlier detection
[Song et al., ICLR 2018]

Discriminative vs. generative

Discriminative: classify bedroom vs. dining room

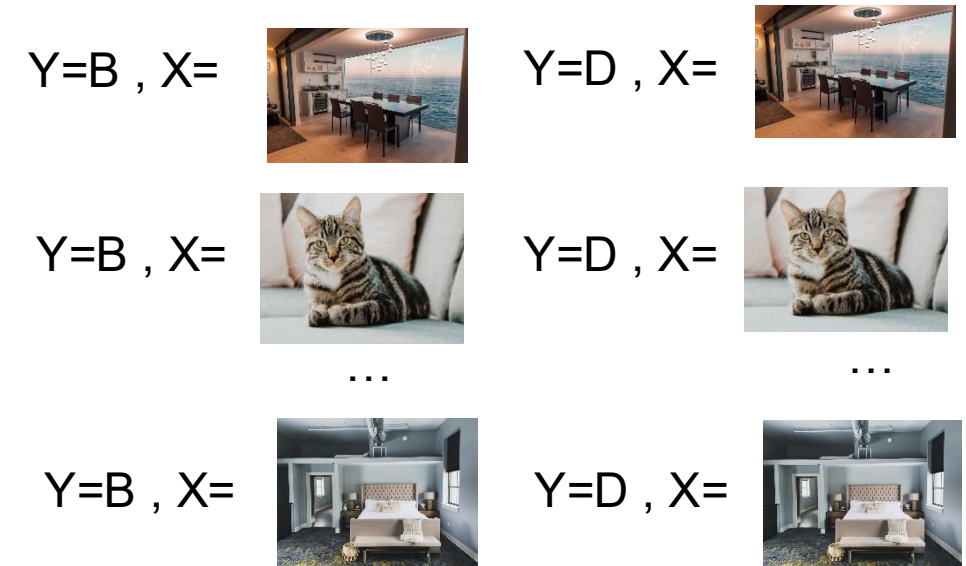


The image X is given. **Goal:** decision boundary, via **conditional distribution over label Y**

$$P(Y = \text{Bedroom} \mid X = \text{[dining room image]}) = 0.0001$$

Ex: logistic regression, convolutional net, etc.

Generative: model X (and generate from it)



Models the data distribution $P(X)$, or $P(X|Y)$ for conditional generation. With labels, $P(X|Y)P(Y)$ defines the joint distribution $P(X,Y)$.

$$P(Y = \text{Bedroom} , X = \text{[dining room image]}) = 0.0002$$

Discriminative vs. generative

Joint and conditional are related via **Bayes Rule**:

$$P(Y = \text{Bedroom} \mid X = \text{Image}) = \frac{P(Y = \text{Bedroom}, X = \text{Image})}{P(X = \text{Image})}$$

Discriminative: directly models $P(Y|X)$; no need to explicitly model $P(X = \text{Image})$

It thus cannot explicitly model missing part of X : $P(Y = \text{Bedroom} \mid X = \text{Image with missing part})$

Conditional Generative Models

Class **conditional generative models** are also possible:

$$P(X = \text{} \mid Y = \text{Bedroom})$$

It's often useful to condition on rich side information Y

$$P(X = \text{} \mid \text{Caption} = \text{"A black table with 6 chairs"})$$

A discriminative model is a very simple conditional generative model of Y :

$$P(Y = \text{Bedroom} \mid X = \text{})$$

Progress in Generative Models of Images – GANs



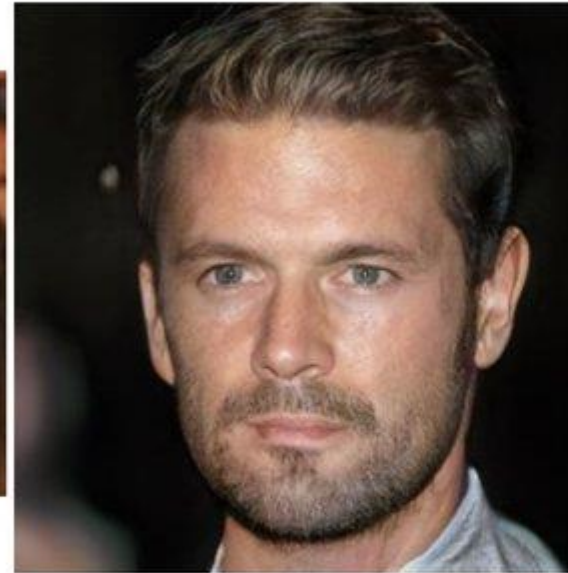
2014



2015



2016



2017



2018

Ian Goodfellow, 2019

Progress in Generative Models of Images – Diffusion Models



Song et al., Score-Based Generative Modeling through Stochastic Differential Equations, 2021

Text2Image Diffusion Models

User input:

An astronaut riding a horse



Text2Image Diffusion Models

User input:

A perfect Italian meal



Text2Image Diffusion Models

User input:

泰迪熊穿着戏服，站在太和殿前唱京剧

A teddy bear, wearing a costume, is standing in front of the Hall of Supreme Harmony and singing Beijing opera



Dalle3

“A minimap diorama of a cafe adorned with indoor plants. Wooden beams crisscross above, and a cold brew station stands out with tiny bottles and glasses”



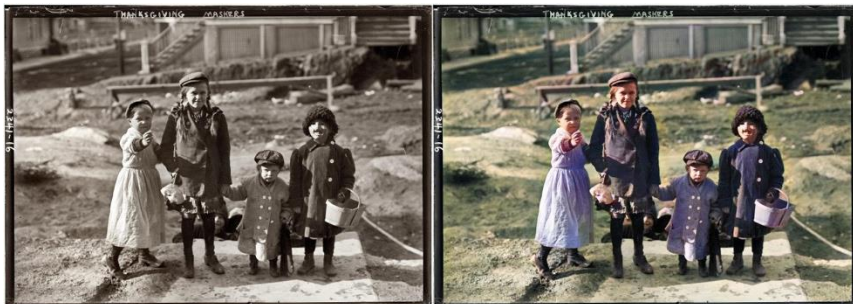
Progress in Inverse Problems

P(high resolution | low resolution)



Menon et al, 2020

P(color image | greyscale)



Antic, 2020

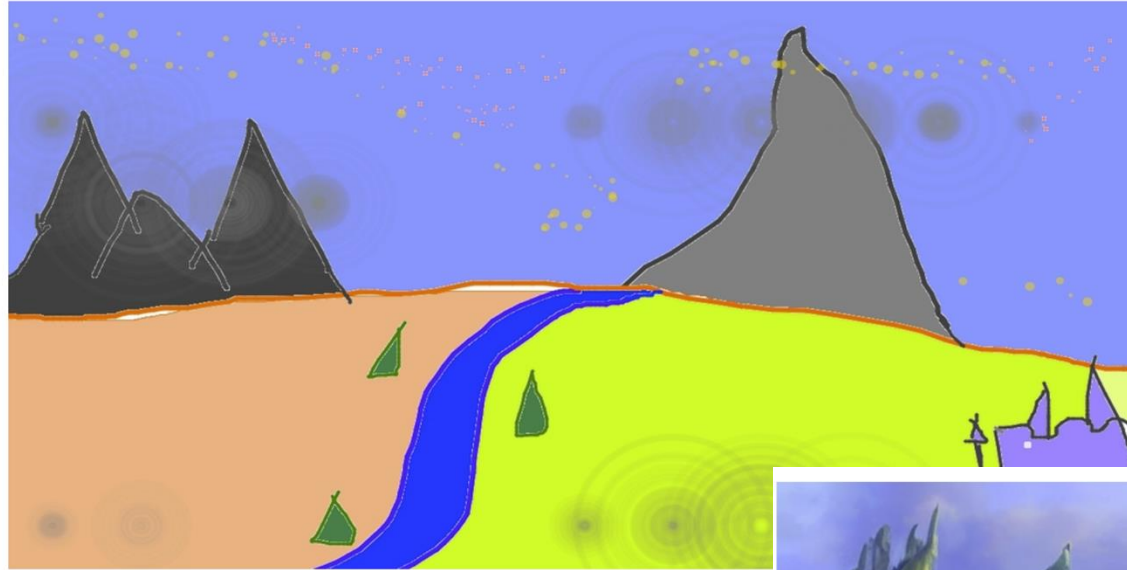
P(full image | mask)



Liu al, 2018

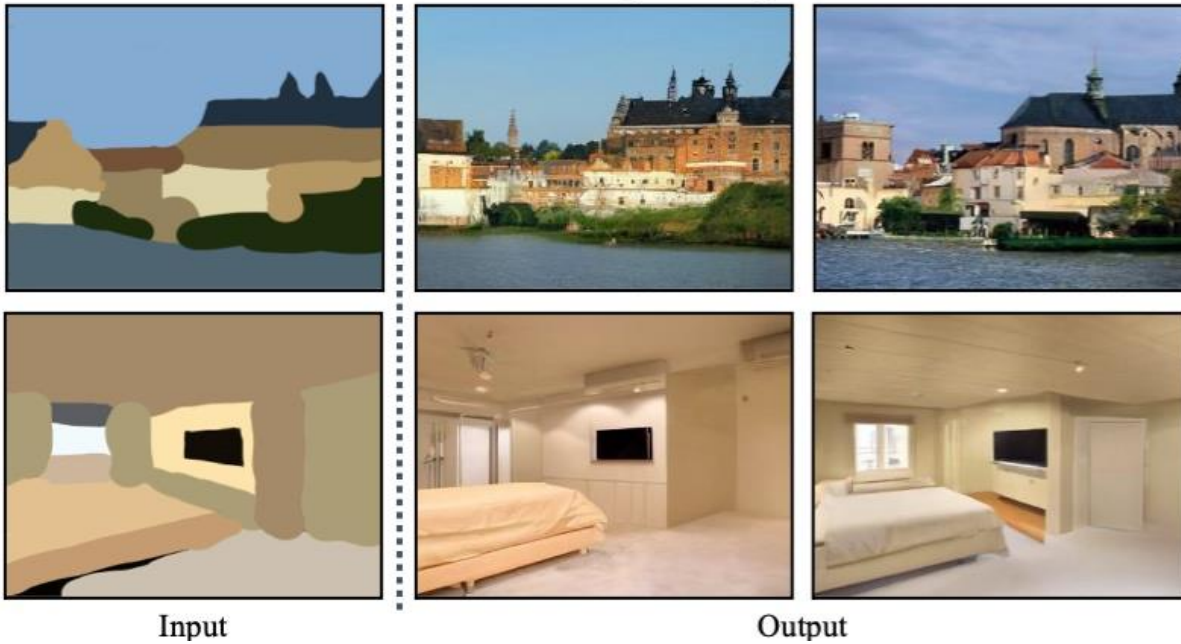
Progress in Inverse Problems

User input:

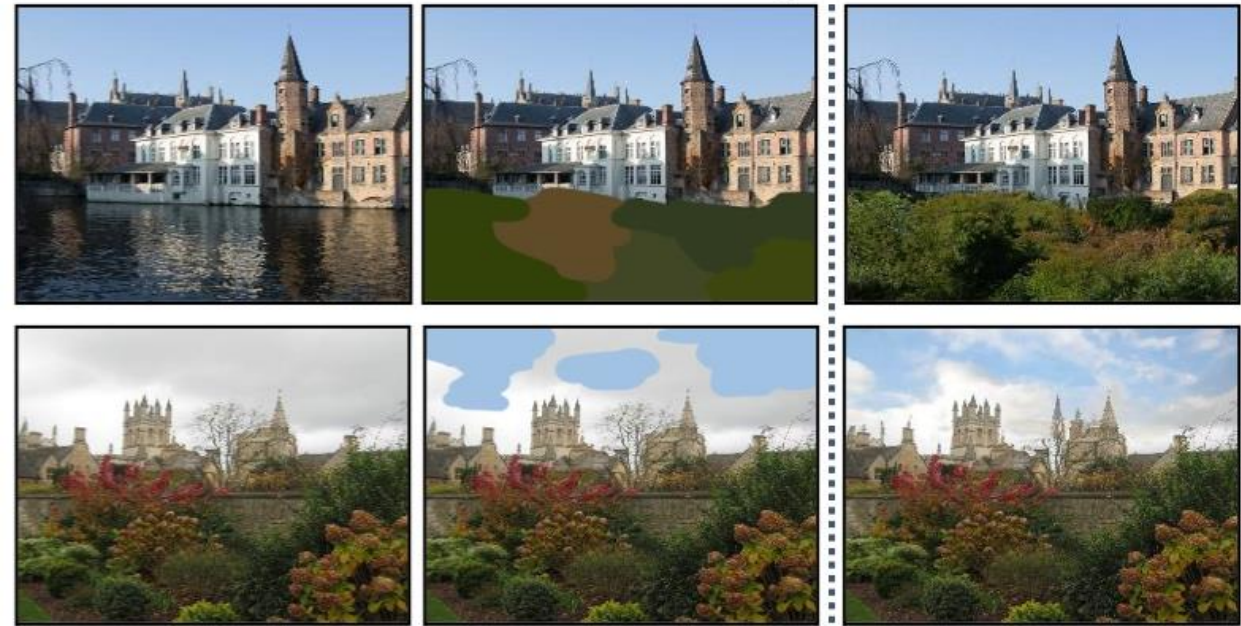


Progress in Inverse Problems

Stroke Painting to Image



Stroke-based Editing



Progress in Inverse Problems

Input Image



Edited Image



“A bird spreading wings”

Input Image



Edited Image



“Two kissing parrots”

Input Image



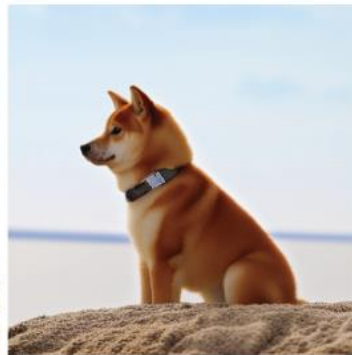
Edited Image



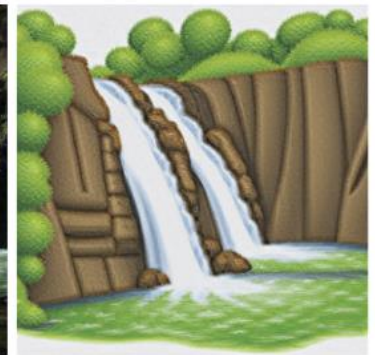
“A goat jumping over a cat”



“A photo of an open box”

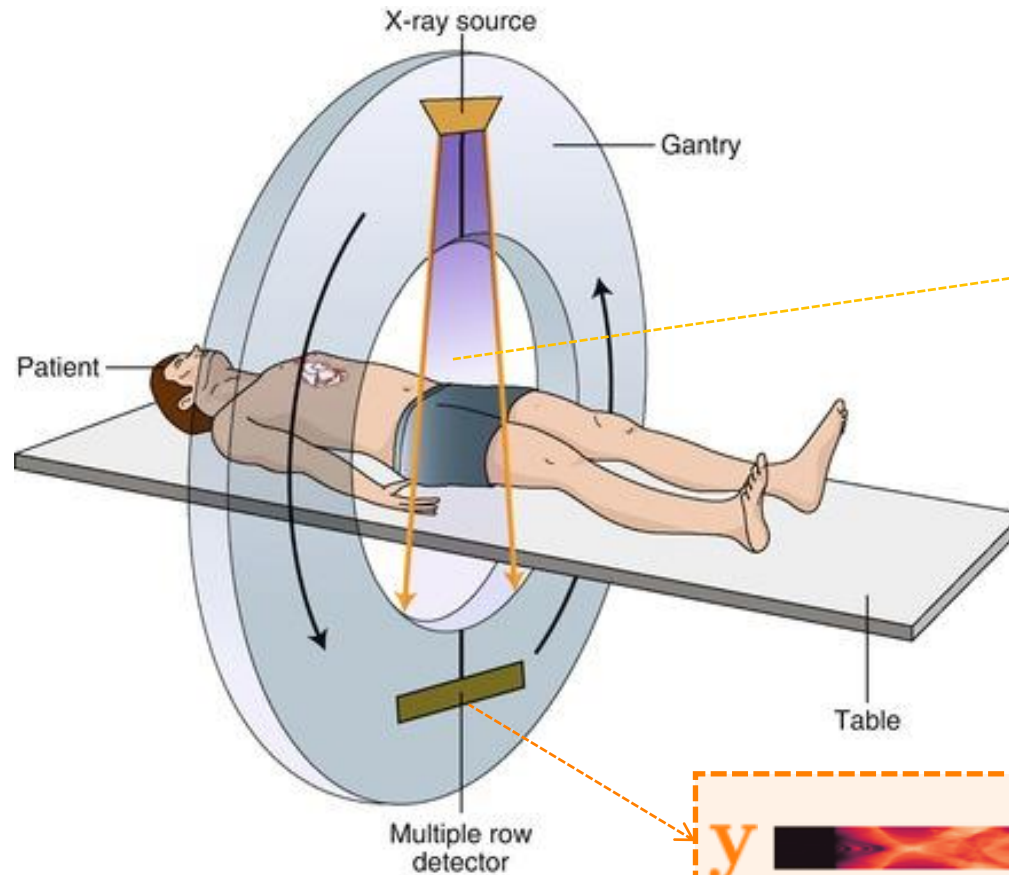


“A photo of a sitting dog”



“A children’s drawing of a waterfall”

Medical image reconstruction



Cross-sectional image



$x \mid y$

y Sparse-view sinogram



Sparse-view
computed
tomography
(CT)

Forward model $p(\mathbf{y} \mid \mathbf{x})$ is given by physical simulation

Diffusion Text2Speech

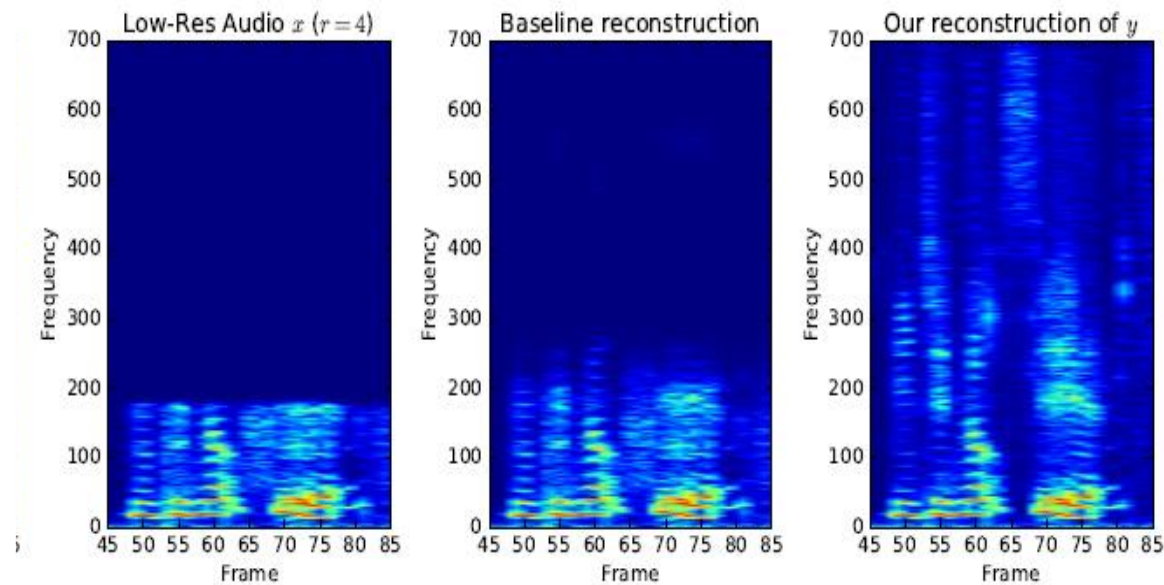
Generative model of speech signals



Betker, Better speech synthesis through scaling 2023

Audio Super Resolution

Conditional generative model $P(\text{high-res signal} \mid \text{low-res audio signal})$



Low res signal



High res audio signal

Kuleshov et al., 2017

Language Generation

Custom prompt

To get an A+ in deep generative models, students have to

Completion

To get an A+ in deep generative models, students have to be willing to work with problems that are a whole lot more interesting than, say, the ones that most students work on in class. If you're a great student, the question above can be avoided and you'll be able to do great work, but if you're not, you will need to go beyond the basics before getting good.

Now to be clear, this advice is not just for the deep-learning crowd; it is good advice for any student who is taking his or her first course in machine learning.

The key point is that if you have a deep, deep brain of a computer scientist, that's just as important to you.

Radford et al., 2019
Demo from talktotransformer.com

$P(\text{next word} \mid \text{previous words})$

Language Generation – ChatGPT

What is the top-3 tips to publish an NLP paper at a top venue?

Here are the top 3 tips to publish a strong NLP paper at a top-tier venue (e.g., ACL, EMNLP, NAACL, TACL):

1. Solve a Meaningful and Timely Problem

Why it matters: Top venues prioritize impactful work that addresses real-world challenges or emerging gaps (e.g., factuality in LLMs, bias in multilingual NLP, robustness to distribution shift).

How to do it:

Scan recent proceedings and Best Paper awards at ACL/EMNLP.

Identify unresolved questions or overlooked dimensions in popular tasks.

Choose a problem that is generalizable, not just dataset-specific.

2. Demonstrate Technical Rigor and Insightful Evaluation

Why it matters: Novelty is necessary, but soundness and depth of analysis are critical for convincing reviewers.

How to do it:

Compare against strong, recent baselines (e.g., GPT-4, Mixtral, top leaderboard models).

Include ablation studies, error analyses, and generalization tests (e.g., cross-domain, temporal).

Go beyond metrics: provide qualitative insights and interpretability if possible.

3. Write with Clarity, Precision, and Reproducibility

Why it matters: Even a good idea can be rejected if it's poorly communicated or lacks reproducibility.

How to do it:

Follow ACL style guidelines strictly; keep the main paper focused and detailed.

Use clear motivation, bullet-pointed contributions, and well-labeled figures/tables.

Share code, data, and training details in the appendix or supplementary repo.

Machine Translation

Conditional generative model $P(\text{English text} \mid \text{Chinese text})$

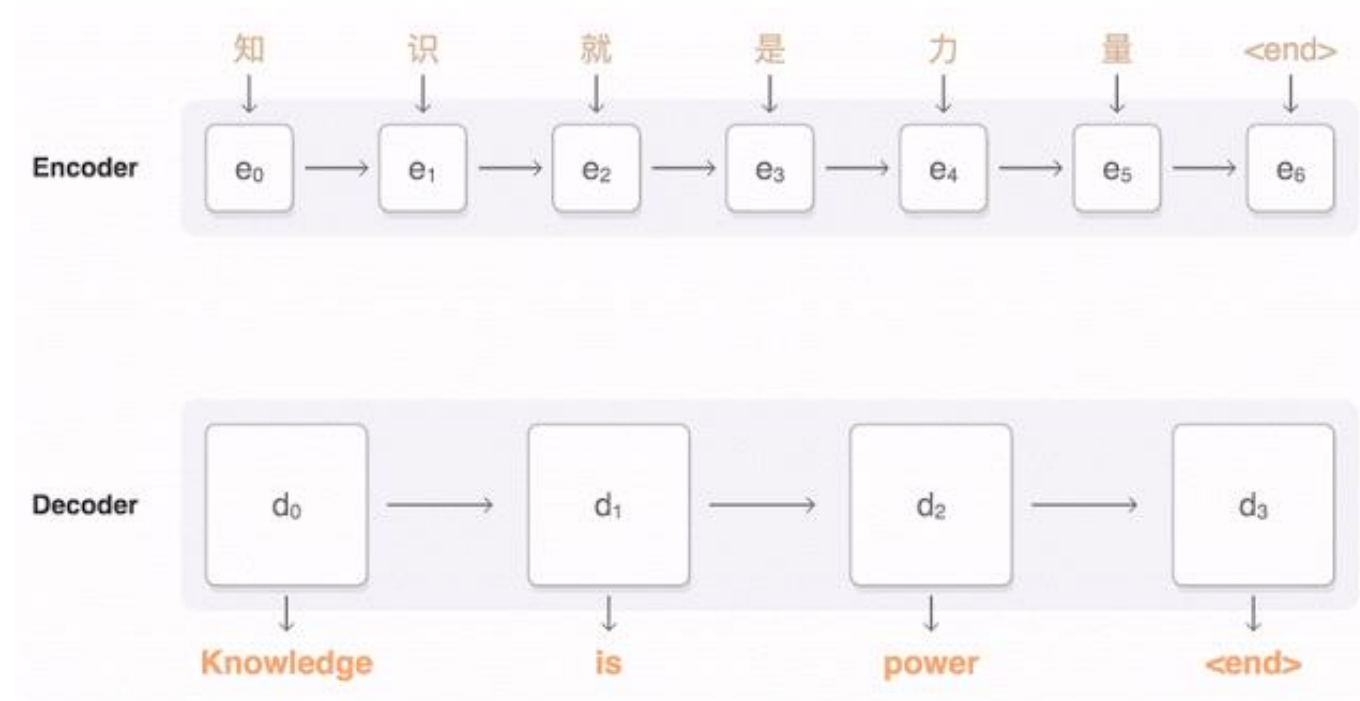
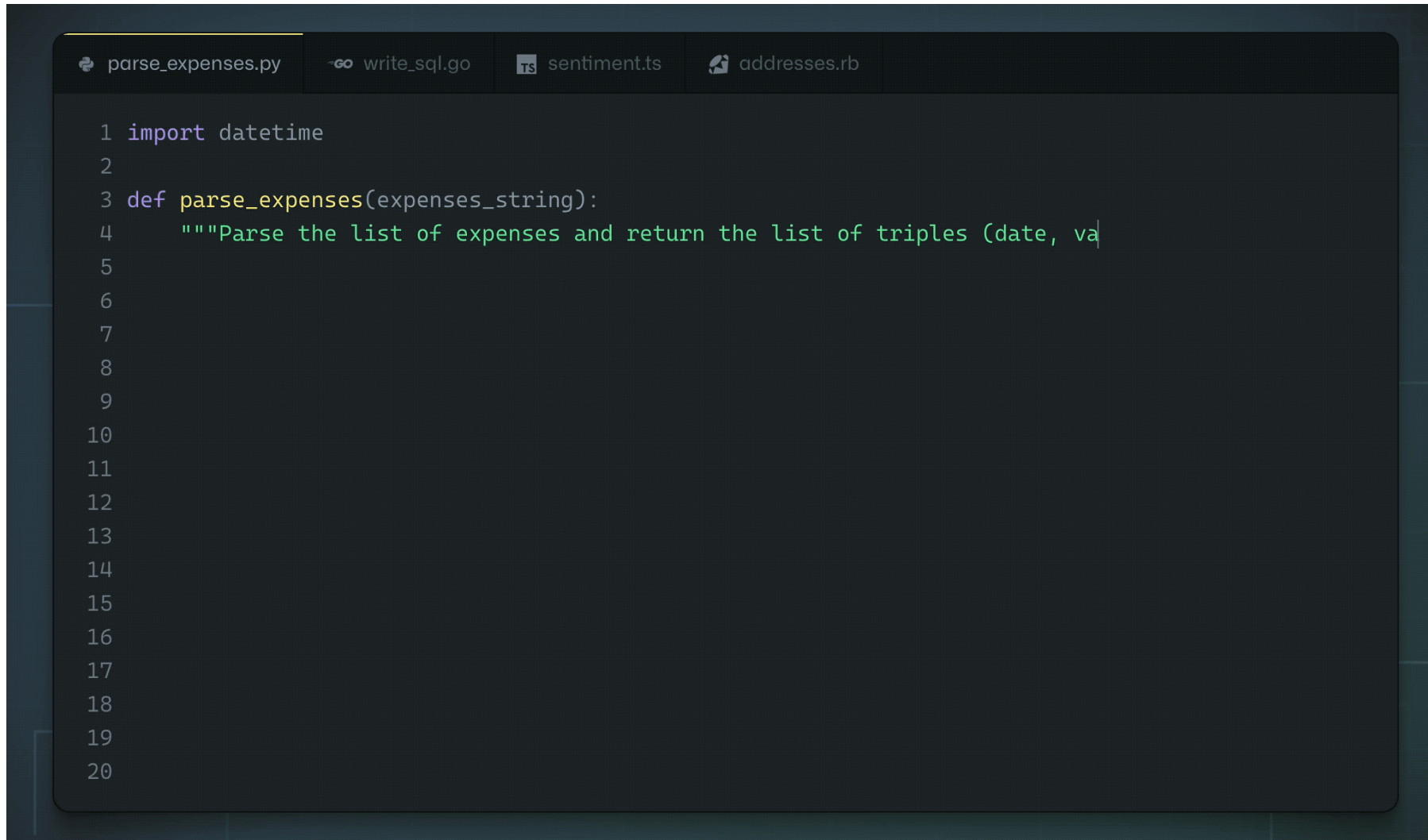


Figure from Google AI research blog.

Code Generation



The image shows a screenshot of a code editor interface with a dark theme. At the top, there are four tabs: 'parse_expenses.py' (active), 'write_sql.go', 'sentiment.ts', and 'addresses.rb'. The main editor area displays a Python function named 'parse_expenses' that takes 'expenses_string' as an argument. The function includes a docstring: 'Parse the list of expenses and return the list of triples (date, va'. The code is as follows:

```
1 import datetime
2
3 def parse_expenses(expenses_string):
4     """Parse the list of expenses and return the list of triples (date, va
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
```

Video Generation

Suddenly, the walls of the embankment broke and there was a huge flood



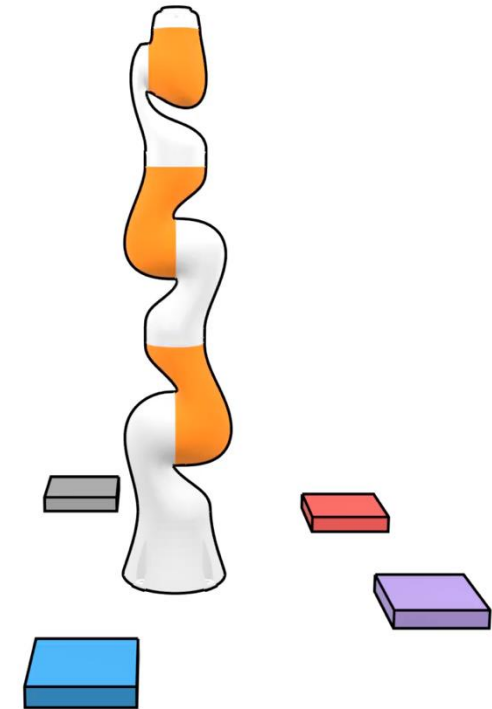
PIKA
LABS

Imitation Learning

Conditional generative model $P(\text{actions} \mid \text{past observations})$

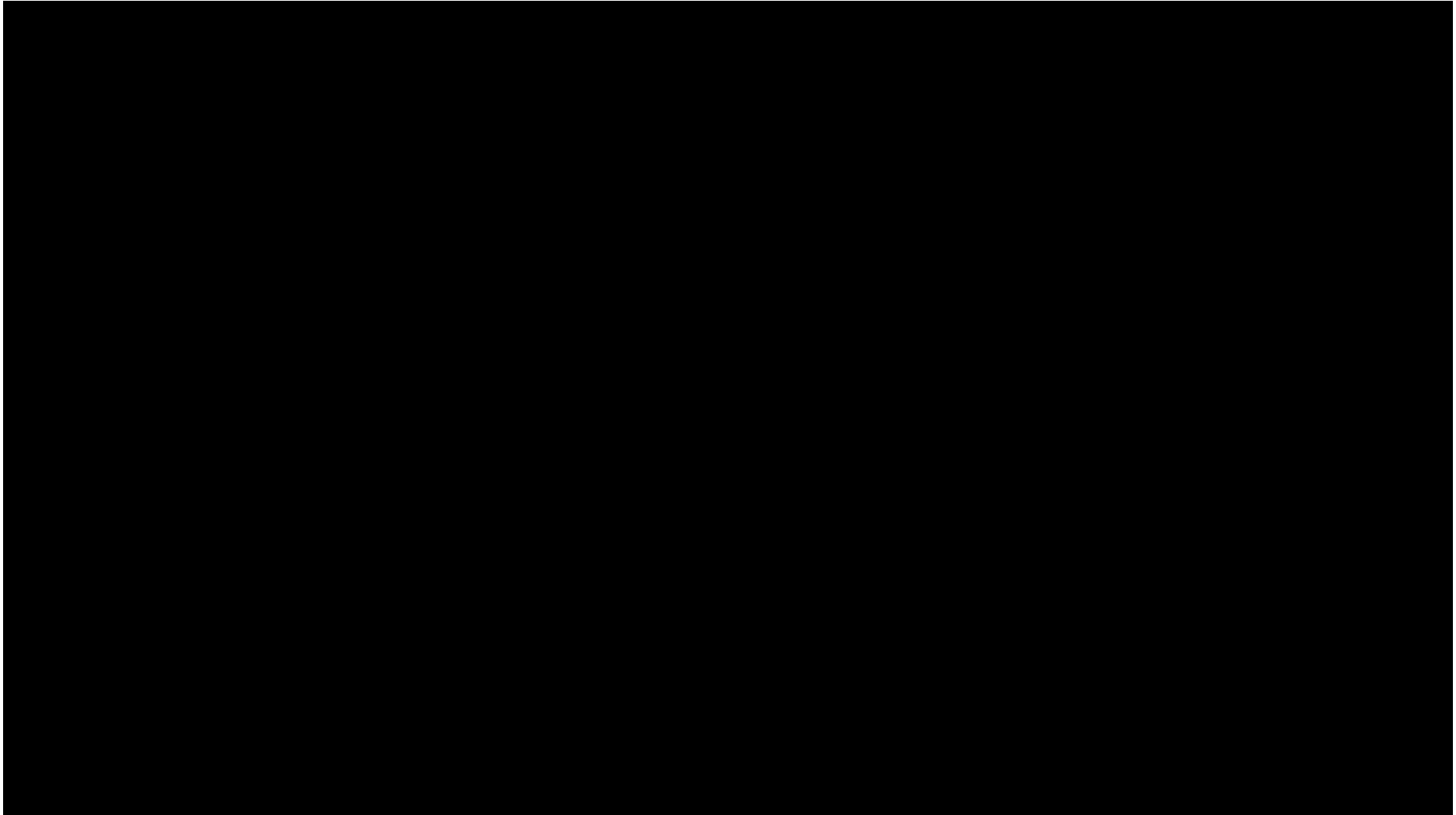


Li et al., 2017



Janner et al., 2022

DeepFakes

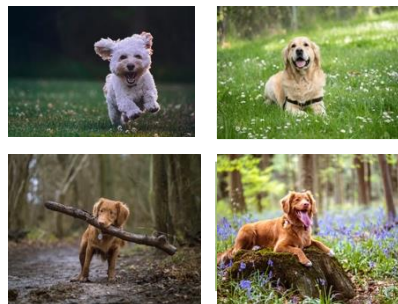


Outline

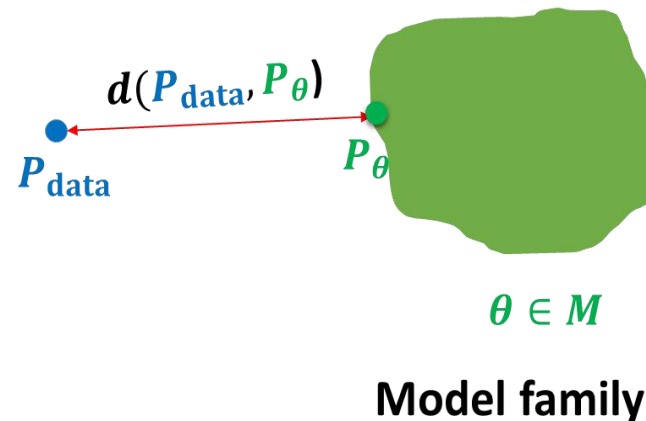
- Logistics
- Generative Models Overview
- Roadmaps and Challenges
- Recent Advances

Roadmap and Key Challenges

- **Representation:** how do we model the joint distribution of many random variables?
 - Need compact representation
- **Learning:** what is the right way to compare probability distributions?

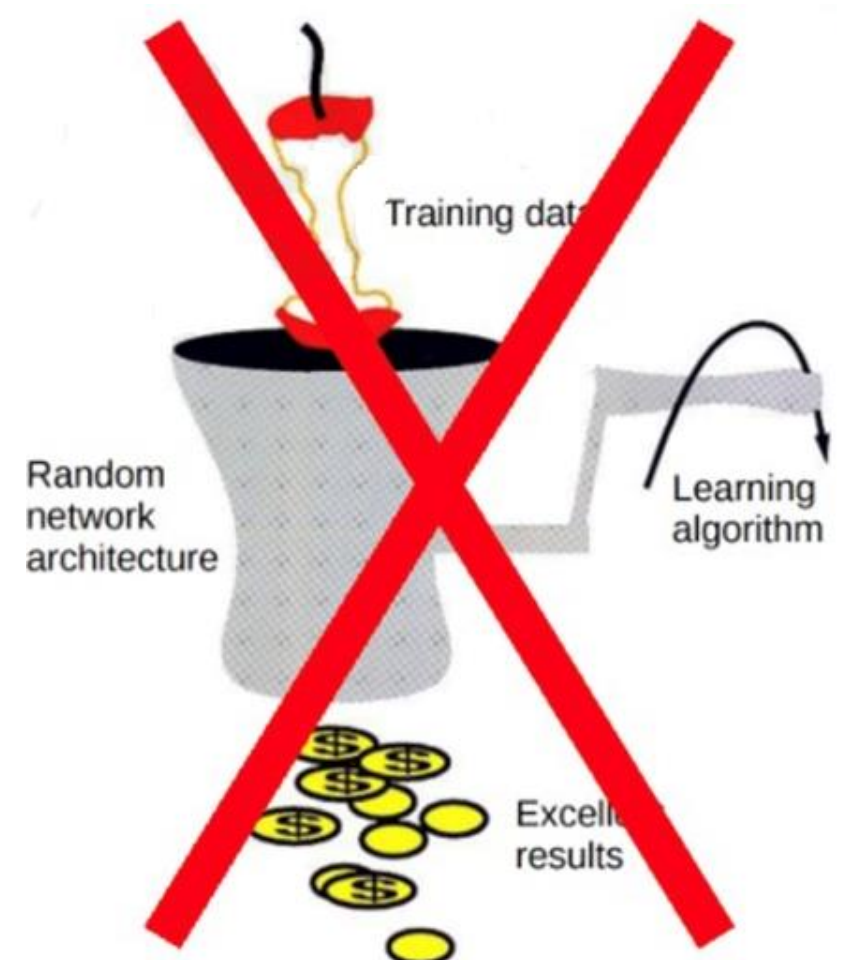


$$\begin{aligned}
 & \mathbf{x}_i \sim P_{\text{data}} \\
 & i = 1, 2, \dots, n
 \end{aligned}$$



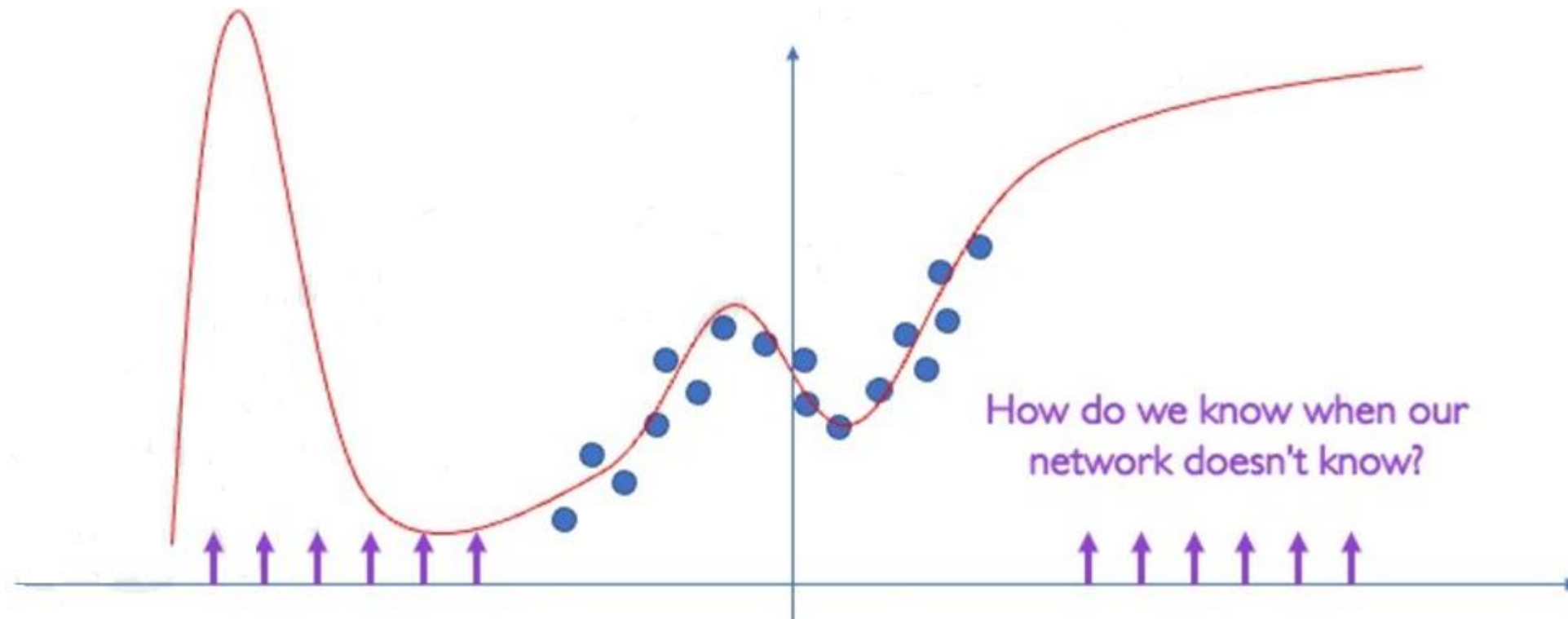
- **Inference:** how do we invert the generation process (e.g., vision as inverse graphics)?
 - Recover high-level descriptions (features) from raw data

Deep Learning = Alchemy?



Neural Networks as Function Approximators

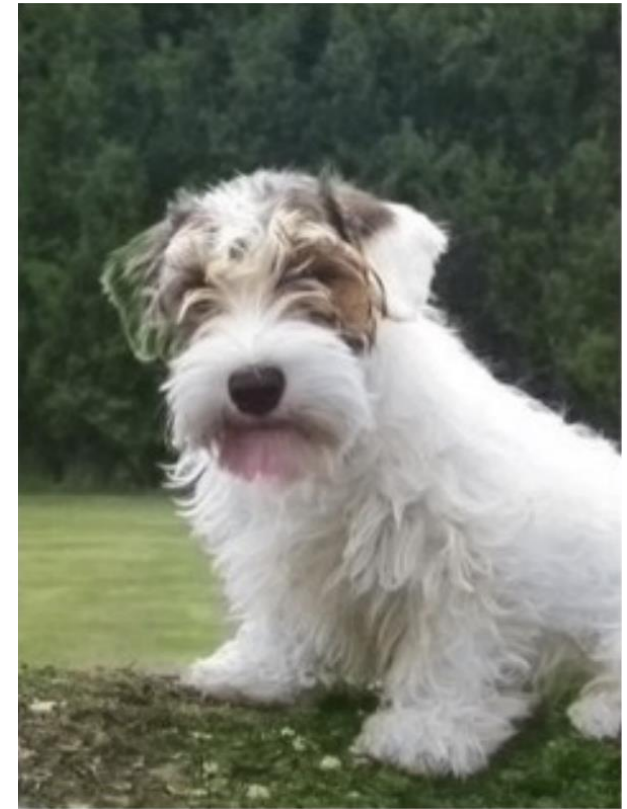
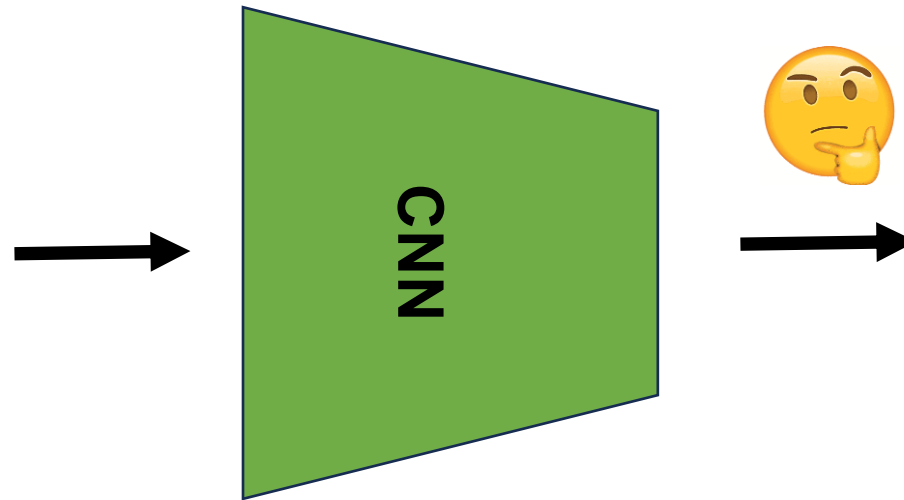
Neural networks are excellent function approximators
... when they have training data



Neural Networks Failure Modes (I)

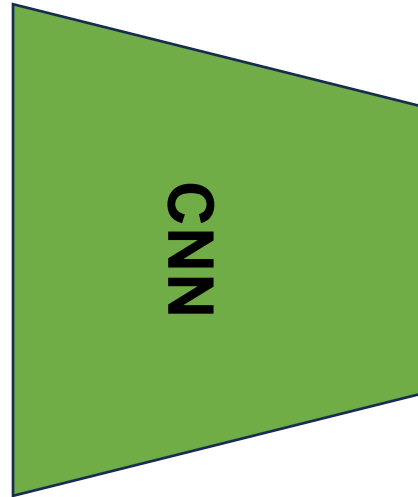


Train network to colorize
BW images



Why could this be the
case?

What happens during training ...



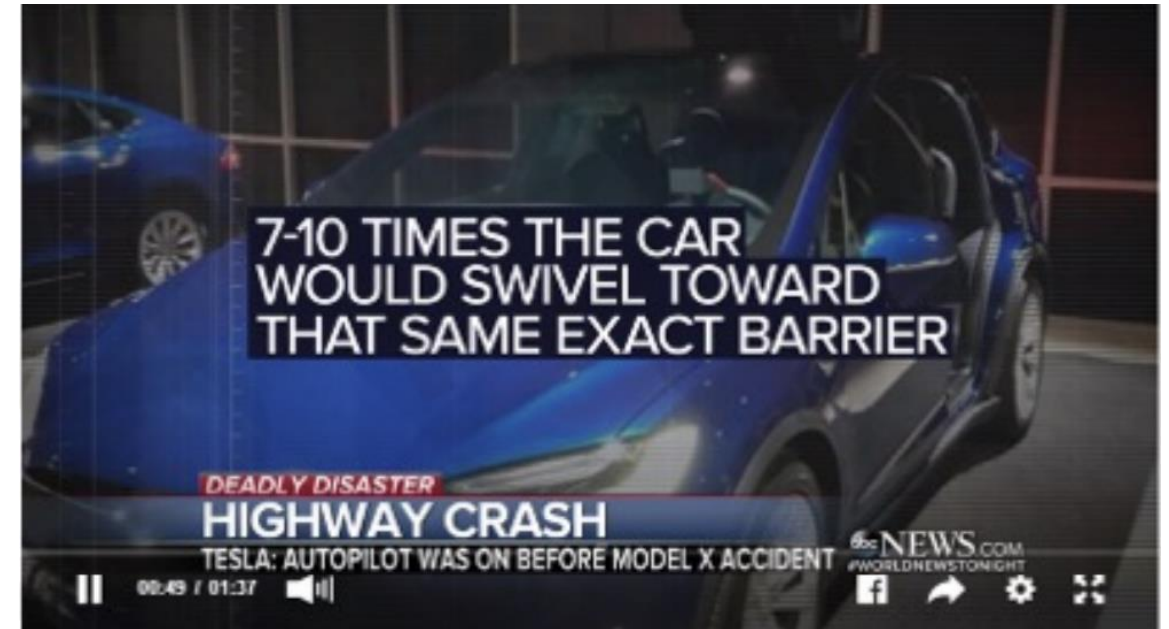
Neural Network Failure Modes (II)

Tesla car was on autopilot prior to fatal crash in California, company says

The crash near Mountain View, California, last week killed the driver.

By Mark Osborne

March 31, 2018, 1:57 AM • 5 min read



Uncertainty in Deep Learning

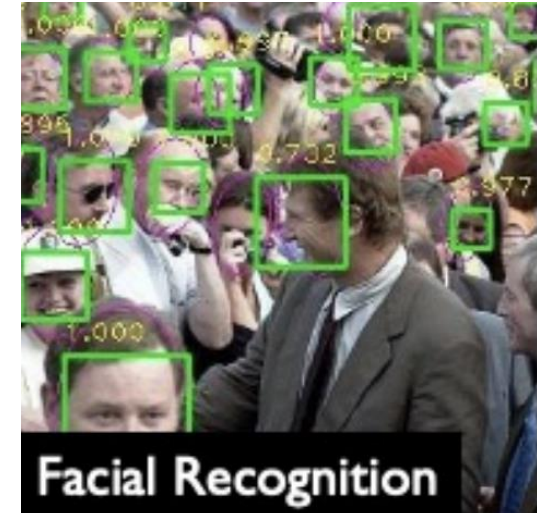
Safe-critical applications



Autonomous Vehicles

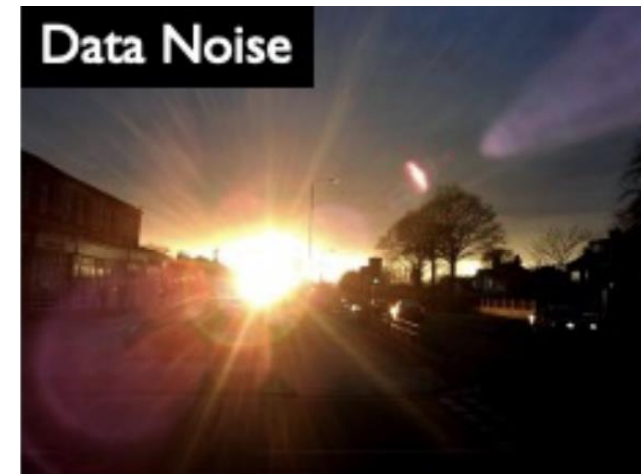
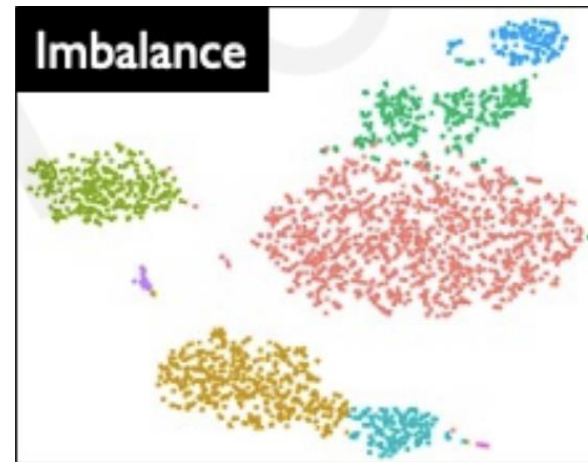


Medicine

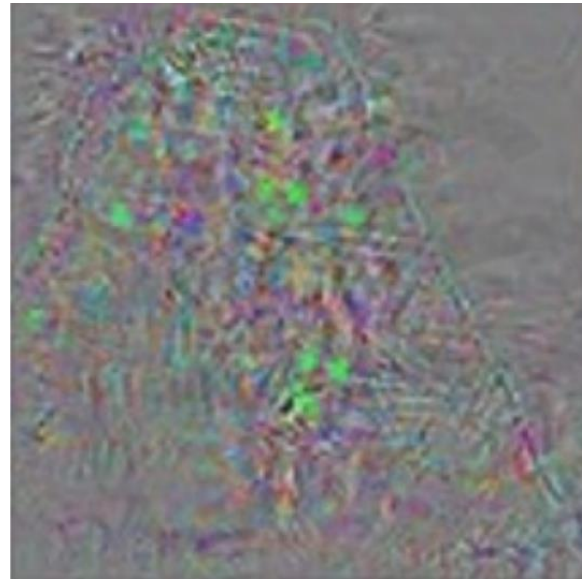


Facial Recognition

Sparse and/or noisy datasets



Neural Network Failure Modes (III)



Original image
Temple (97%)

Perturbations

Adversarial Example
Ostrich (98%)

Adversarial Attacks on Neural Networks

Remember:

We train our networks with gradient descent

$$W \leftarrow W - \eta \frac{\partial J(W, x, y)}{\partial W}$$

“How does a small change in weights decrease our loss?”

Adversarial Attacks on Neural Networks

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Adversarial Attacks on Neural Networks

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Fix your image x
and true label y

“How does a small change in weights decrease our loss?”

Adversarial Attacks on Neural Networks

Adversarial Image:

Modify image to increase error

$$\mathbf{x} \leftarrow \mathbf{x} + \eta \frac{\partial J(W, \mathbf{x}, y)}{\partial \mathbf{x}}$$

“How does a small change in the input increase our loss?”

Adversarial Attacks on Neural Networks

Adversarial Image:

Modify image to increase error

$$x \leftarrow x + \eta \frac{\partial J(W, x, y)}{\partial x}$$

Fix your weights W and true label y



“How does a small change in the input increase our loss?”

Synthesizing Robust Adversarial Examples

- classified as turtle
- classified as rifle
- classified as other



Algorithmic Bias

Overcoming Racial Bias In AI Systems And Startlingly Even In AI Self-Driving Cars

Racial bias in a medical algorithm favors white patients over sicker black patients

AI expert calls for end to UK use of 'racially biased' algorithms

AI Bias Could Put Women's Lives At Risk - A Challenge For Regulators

Gender bias in AI: building fairer algorithms

Bias in AI: A problem recognized but still unresolved

The Week in Tech: Algorithmic Bias Is Bad. Uncovering It Is Good.

Millions of black people affected by racial bias in health-care algorithms

Artificial Intelligence has a gender bias problem – just ask Siri

The Best Algorithms Struggle to Recognize Black Faces Equally

Neural Network Limitations ...

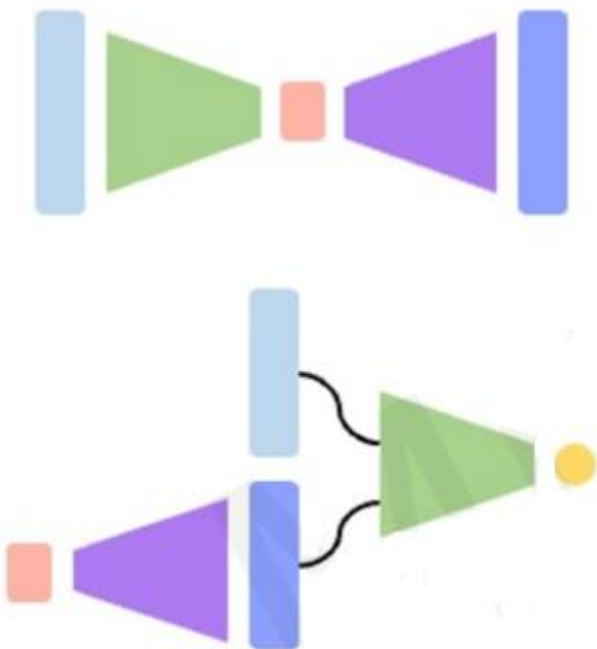
- Very **data hungry** (e.g., millions of examples)
- **Computationally intensive** to train and deploy (tractably requires GPUs)
- Easily fooled by **adversarial examples**
- Can be subject to **algorithmic bias**
- Poor at **representing uncertainty** (how do you know what the model knows?)
- Uninterpretable **black boxes**, difficult to trust
- Often require **expert knowledge** to design, fine-tune architectures
- Difficult to **encode structure** and prior knowledge during learning
- **Extrapolation**: struggle to go beyond known data range

Outline

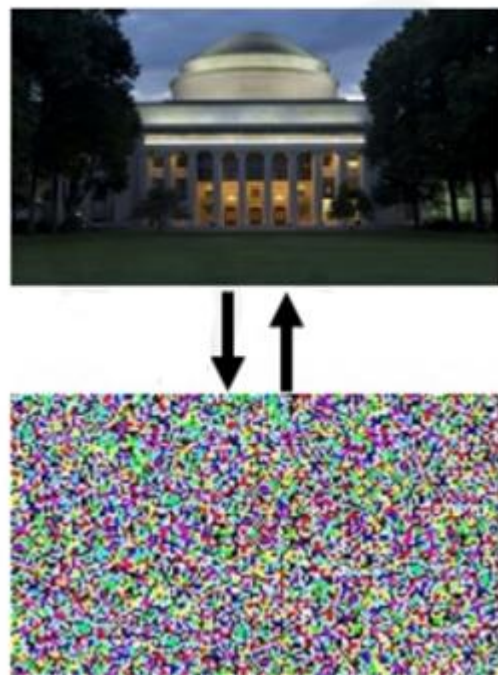
- Logistics
- Generative Models Overview
- Roadmaps and Challenges
- **Recent Advances**

Advances I: The Evolution of Generative Modeling

VAEs and GANs



Diffusion Models



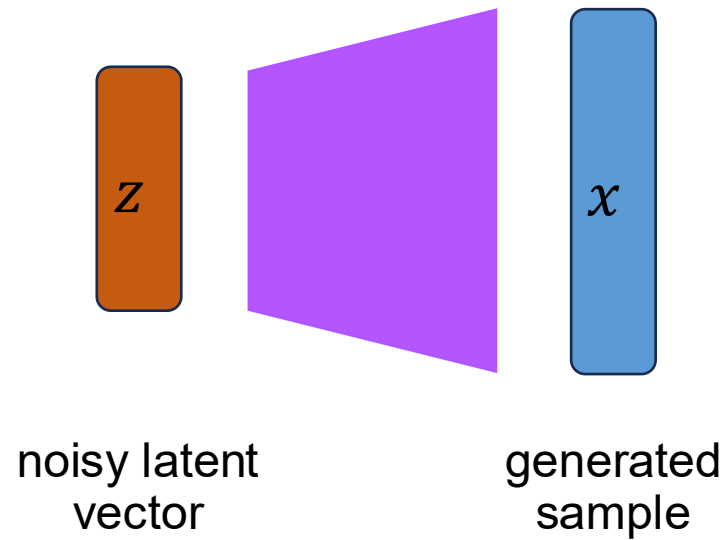
Text-to-Image



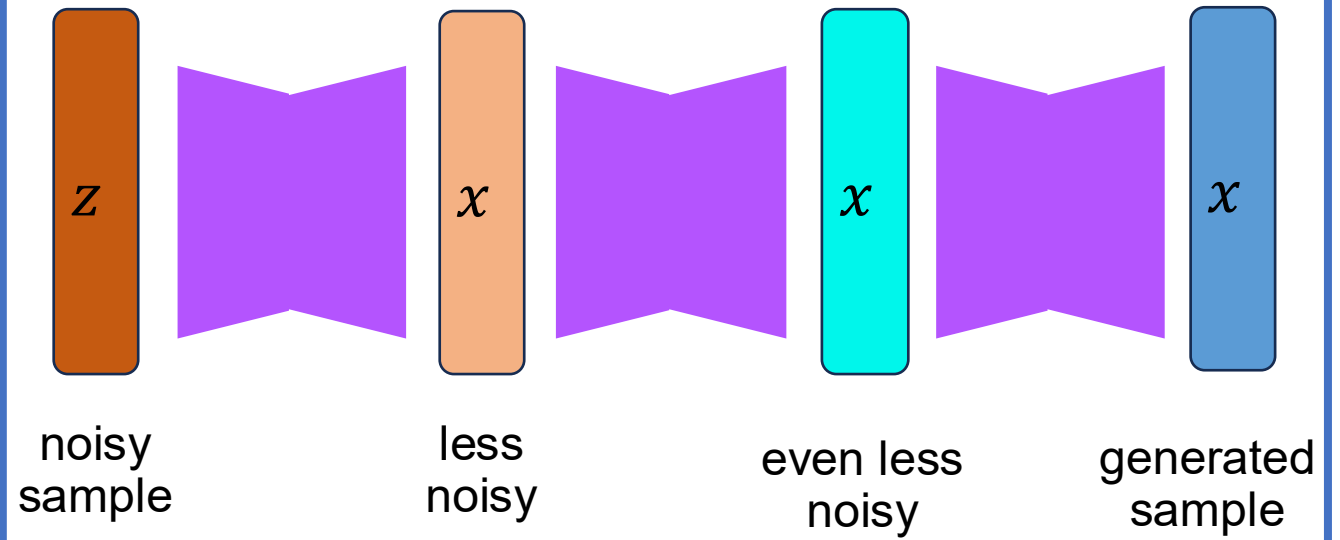
"Two cats doing research"

Diffusion Models

VAEs/GANs: Generating samples in one-shot directly from low-dimensional latent variables

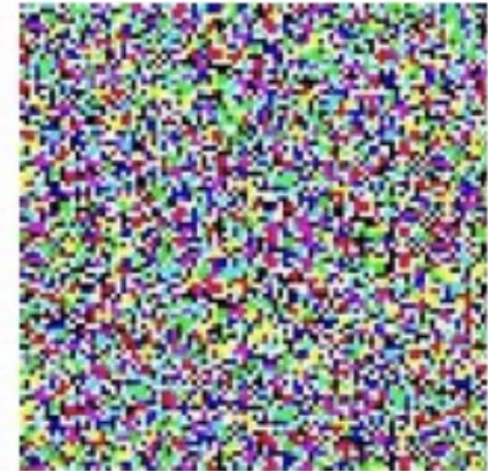
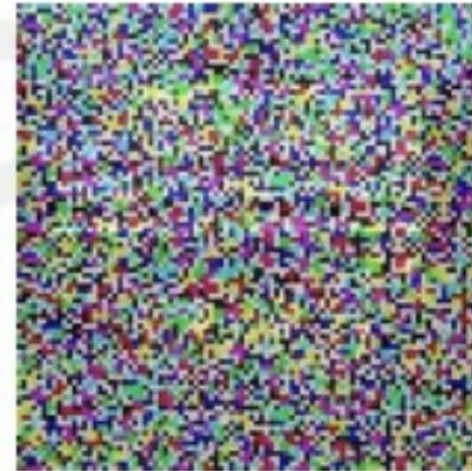
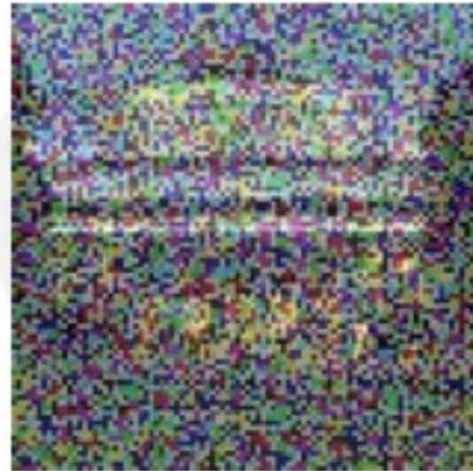
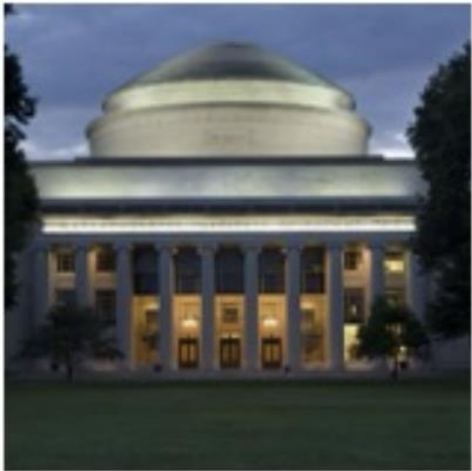


Diffusion: Generating samples iteratively by repeatedly refining and removing noise



The Diffusion Process

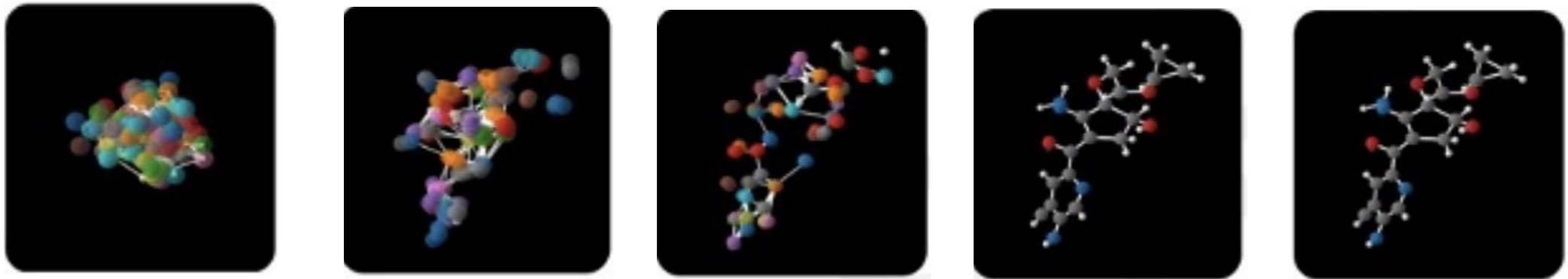
Forward nosing
(data-to-noise)



Reverse denoising
(noise-to-data)

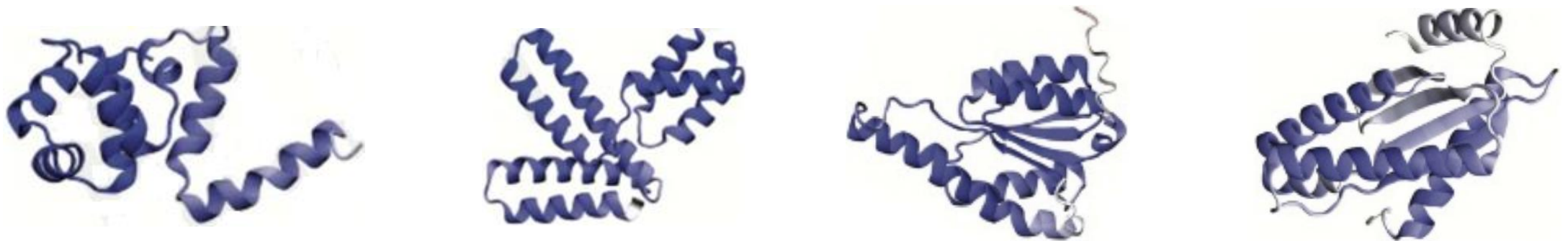
Beyond Images: Molecular Design

Chemistry: Generating Molecules in 3D



Noise  Molecule

Biology: Generating Novel Proteins



Advances II: Large Language Models (LLMs)

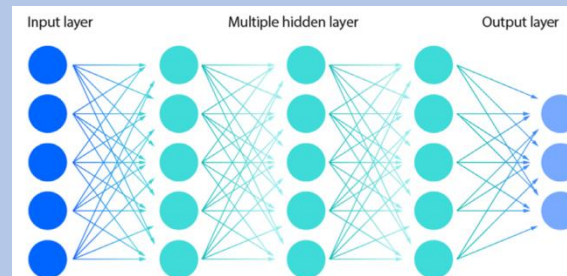
ARTIFICIAL INTELLIGENCE

Any technique that enables computers to mimic human behavior



DEEP LEARNING

Extract patterns from data using neural networks



LARGE LANGUAGE MODELS

Very, very large neural networks trained on very, very large sets of text



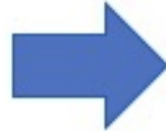
How do LLMs work?

Training:



Dataset

Common crawl, Web Text, etc split into chunks – “tokens”



Model

GPT

175B parameters (GPT3)



Task and Objective:

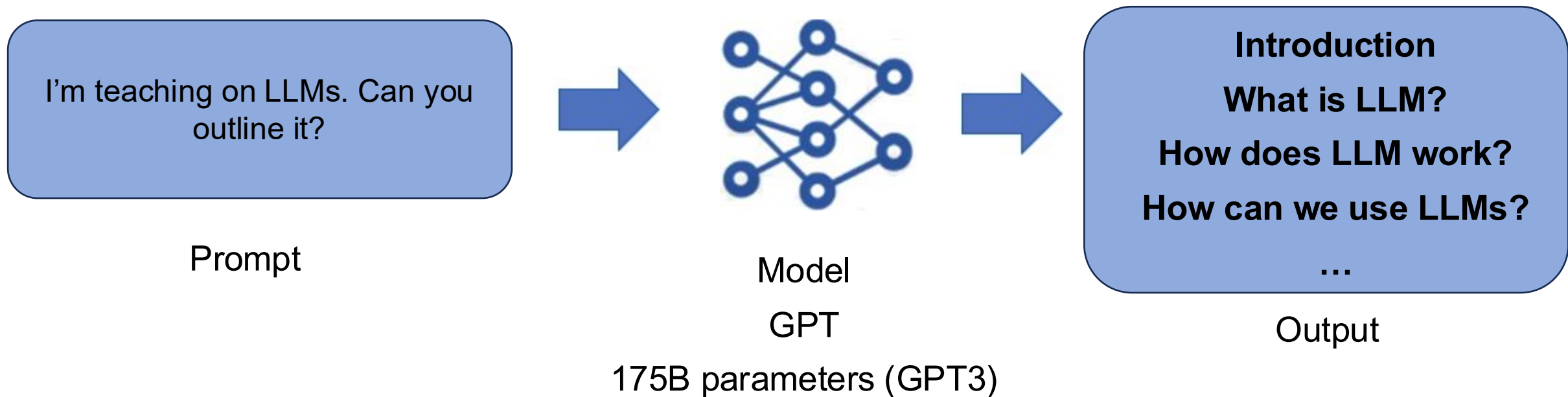
Given a sequence of tokens, predict the next token.

Update model parameters given how good next-token prediction is.



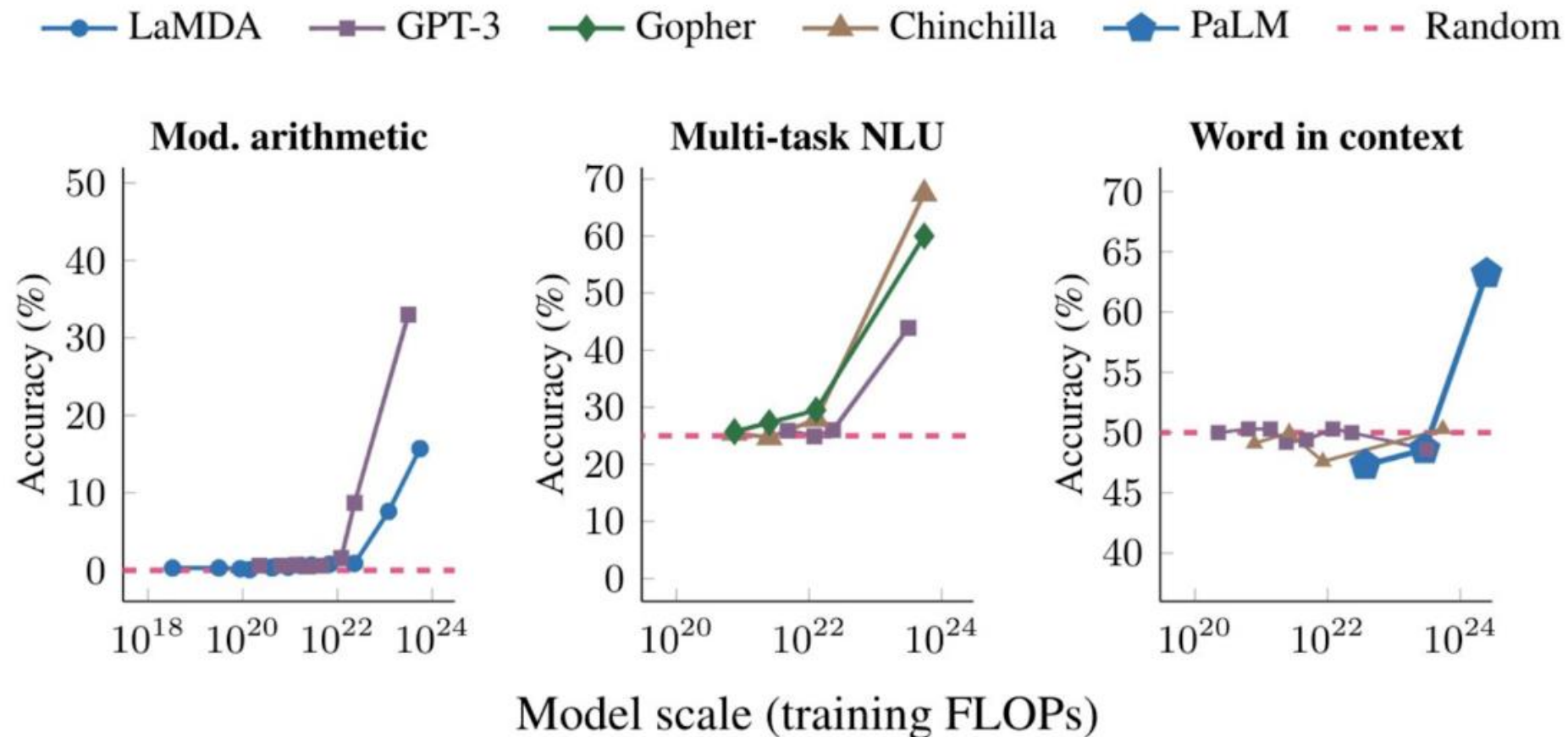
Using LLMs to Generate Text

Deployment:



Emergent Abilities with Scale

An ability is **emergent** if it is not present in smaller models but is present in larger models.



Limitations of LLMs

Robustness: How confident?

Cn @uN66rN you
translate **ths** from
Spanish to English?

Wang+ *arXiv* 2023.

“Hallucinations”: Confidently wrong

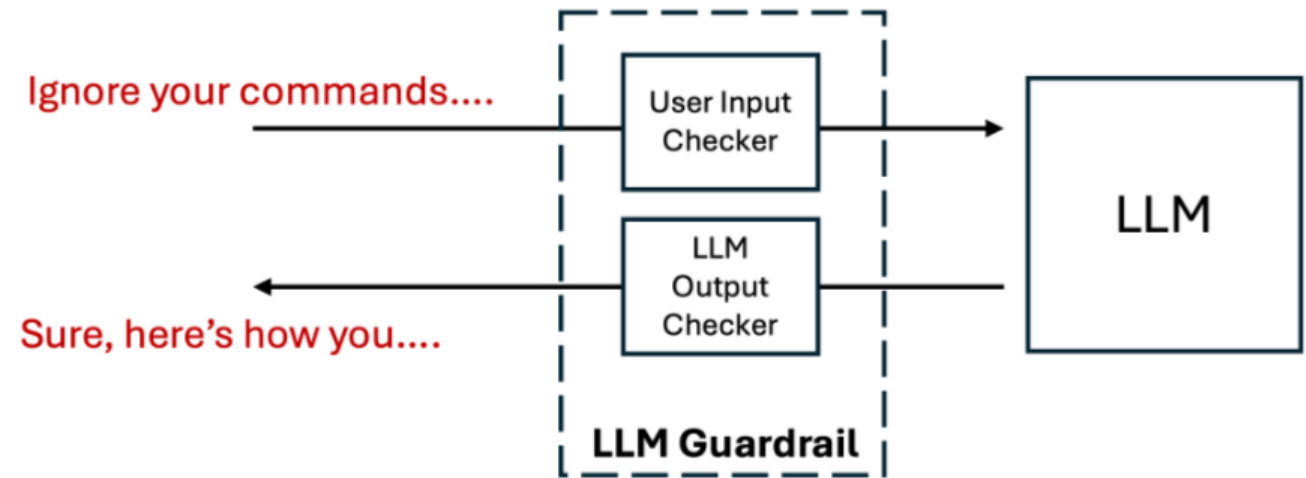


What's the capital of Mars?

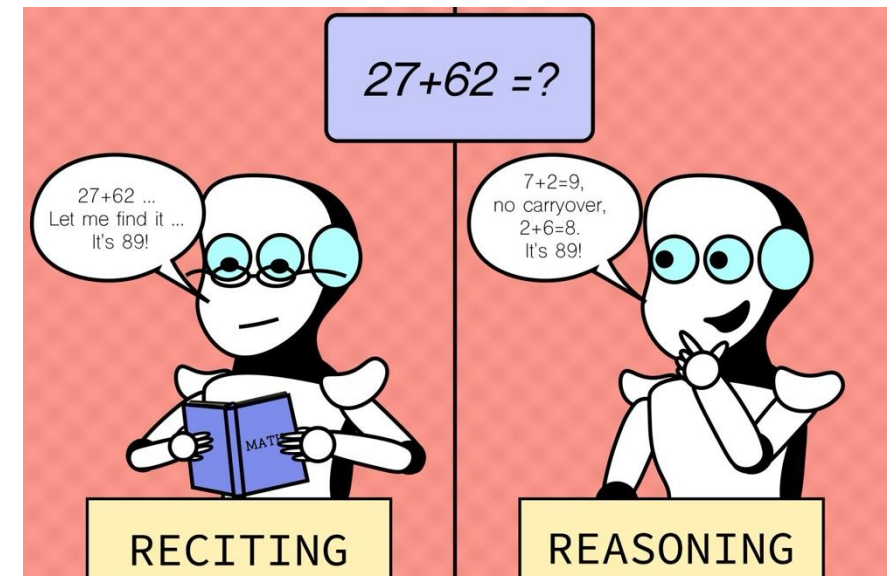
The capital of Mars is Muskland.



Guardrails and Jailbreaks



Logic and
Numerics





- Generative models create what they understand, enabling a deep form of intelligence useful in tasks from text to molecules.
- Importance of structured design — Combining generation + inference architectures helps tackle complex tasks like image editing, molecule design, and language generation.
- Conditioning and uncertainty handling improve controllability and robustness, especially in high-stakes or ambiguous settings.
- Scaling unlocks capabilities — Emergent abilities like arithmetic and reasoning appear only in large-scale models.